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COLLEGE OF ENGINEERING, PUNE
(An autonomous Institute of Government of Maharashtra)

END-SEMESTER EXAMINATION

CT 206 - PRINCIPLES OF PROGRAMMING LANGUAGES

S. Y. B. TECH (COMPUTER ENGINEERING / INFORMATION TECHNOLOGY)

Year: 2012-13
Duration: 3 hrs

Semester: IV
Max. Marks: - 100

Instructions:

1. Solve SECTION A and SECTION B on separate answer sheets.
2. Figures to right indicate full marks.
3. Assume suitable data wherever necessary.
4. Draw neat diagrams wherever required.
5. Justify your answer giving appropriate examples.

SECTION A

- Q.1 a) Write a lambda function in Lisp to calculate factorial of a number. Demonstrate use of function. [4]

OR

- a) Write down function to get fourth element of a list in Lisp. Demonstrate use of function. [4]

- b) Write a prolog database for the scenario given below : [4]

- Prof. Turing takes course CT9020.
- Prof. Codd takes course CT93211 and CT9314.
- Prof. Backus takes course CT9021.
- Prof. Ricthie takes course CT9201.
- Prof. Minsky takes course CT9414.
- Fred studies in fy and lectures in courses CT9020.
- Jack and Jill studies in sy.
- Jack lectures in courses CT9311 while Jill lectures in courses CT9314 and CT9414.
- Hendry studies in final year and lectures in courses CT9414 and CT9314.

Hint : lectures(X, Y): person X lectures in course Y

- c) For the above mentioned database write queries and answers (in Prolog) to determine: [4]
- i. Whether Prof. Turing takes course CT9020 or not.
 - ii. Who studies CT9314?
 - iii. Does Turing teach Fred?
 - iv. Who studies in final year?

- d) The two primitive objects in Little quilt language are the square pieces, shown below. [6]



Lets us name these two pieces as a and b respectively.

The expression for the language is given as :

`<expression> ::= a | b | turn(<expression>) | sew (<expression>, <expression>)`

Where turn is an operation of turning a piece by 90 degree in clockwise direction and sew is joining two pieces a and b adjacently. (right side of a with left side of b)

Draw appropriate diagram of the pieces that you will get after execution of following expressions.

i) `sew( turn(turn(b)), a)`

ii) `turn( turn(a) )`

- Q.2. a) Why cost of use becomes a tricky attribute of a good programming language w.r.t. program execution, translation, testing, maintenance. [8]

- b) What are advantages and disadvantages of dynamic type checking? [8]

- Q.3. a) Write a note on [8]

i. Type conversion.

ii. Type inference

- b) Explain stack based implementation for simple call-return subprogram. [8]

SECTION B

- Q.4. a) What is the use of join() function in multithreaded application. Explain with an example. [4]

- b) What is thread synchronization? When is it needed? Support your answer with an example. [8]

OR

- b) What are the steps involved in thread synchronization? Explain in detail. [8]

- c) Suppose you have developed a multithreaded system for a restaurant. Consider following facts for the system :

- There are 7 chefs available to serve the requests.
- There are only 5 gas burners available for cooking.
- Maximum 50 customers present in the restaurant, each orders a separate dish which is to be cooked on a gas burner.

- i. Which is/are the shared resource(es) of the system? [1]
- ii. What are the different types and number of threads that can be present in the system? [3]
- iii. Explain how synchronization is achieved in this case? [2]

Q. 5.

You have a class representing a BankAccount. The account has

- Various data members e.g. acctNumber, lastAccessDate, acctBalance, acctHolderName.
- All the data members except acctNumber are private.
- After every credit / debit transaction, the acctBalance has to be updated.

Class Transaction has

- Data members like transactionId, transactionType (credit / debit), acctNumber, transactionAmount etc
- All the data members except transactionId are private.
- A method viz. credit() that adds the amount to the available balance of the account while debit() subtracts the amount from the balance.
- debit() should not be allowed if after debit operation, the acctBalance becomes less than Rs. 2000
- After every credit or debit operation, a status has to be printed viz. printStatus() that has following details: acctNumber, acctHolderName, transactionId, transactionType, transactionAmount and acctBalance.

Class RunTransaction has 2 methods :

1. main method that has following:

- 2 objects of class BankAccount i.e. acct1 and acct2 (with suitable data)
- 2 objects of class Transaction i.e. trans1(credit transaction on acct1 with suitable data) and trans2 (debit transaction on acct2 with suitable data)

2. static method getAccount() that takes accountNumber as an argument and returns corresponding BankAccount object.

- a) Explain the class structure with data members and member functions for the above scenario. [10]
- b) What is data hiding? How and where data hiding is achieved in this case w.r.t. Q.5. a) [4]
- c) Explain how the hidden data is accessed in this case w.r.t. Q.5. a) [2]

Q. 6. a) You have a program that :

- Opens a particular file file1.txt for reading which is located at a particular location in a file system e.g. /usr/fs/
- The file is supposed to have 10 different numbers of type integer listed one below the other.
- The program reads the numbers in the file one by one. And stores them in an integer array of size 10
- Opens a particular file file2.txt for writing which is located at a particular location in a file system e.g. /usr/fs/
- Divides a number viz. 10000 by the each number in an array and
- Write the result in file2.txt

i. What are exceptions? How they are handled?

[2]

ii. List at least 6 possible exceptions that can arrive while executing above program.

[6]

b) An object oriented system is developed for Library.

The system has classes inherited from class Book i.e. TextBook, ReferenceBook, Journals, Periodicals etc. Library has various departments inherited from class Department. e.g. Purchase, Cataloguing, Issue etc. Also library has its own staff inherited from class Staff e.g. Librarian, Peon etc.

i. What is the use of packages?

[2]

ii. Explain the package structure for the Library system mentioned above and how each class in this package structure is accessed

[6]